TASK 1: ABS WHEEL SPEED SENSOR DIAGNOSTIC INVESTIGATION

Autocare L2 | Block 8 | Apprentice instructions | v1.1

Task reference: Block_8_Task_1_ABS_Wheel_Speed_Diagnostic

Task overview

Block
Block 8 - Advanced Diagnostics and Workshop Management

Weeks
Weeks 71-80

Duties covered
1-10

KSBs covered
K15-K21, S10-S15, B1, B2

Estimated duration
60 minutes

Assessment type
Practical / knowledge / documentation

Scenario

A vehicle has an ABS or traction warning lamp. You must carry out a structured inspection and test of the wheel speed sensor circuit or output, record evidence and explain the likely fault or referral route.

Objectives

- Carry out safe ABS diagnostic checks.
- Identify ABS sensor type and relevant components.
- Record DTCs, signal outputs or visual findings.
- Explain how the evidence supports a diagnosis.
- Identify effects on related systems and customer communication.

Materials you need at the workstation

- Light vehicle or ABS training rig
- Scan tool, multimeter or oscilloscope if available
- Manufacturer technical information
- PPE and Apprentice Assessment Record

Complete all relevant fields. Enter N/A where a field does not apply.

TASK 1: BLOCK 8 APPRENTICE ASSESSMENT

Practical task instructions and apprentice assessment record

Practical task instructions

1. Prepare the vehicle/rig and confirm the warning lamp or fault condition.
2. Read and record relevant DTCs if equipment is available.
3. Inspect the wheel speed sensor, wiring and reluctor/tone ring as instructed.
4. Check output using a suitable method and record results.
5. Explain findings and reinstate the vehicle/workstation.

Vehicle or equipment details

Vehicle make

Mileage / hours

Vehicle model

Engine size / system type

Registration / fleet number

VIN / chassis / equipment ID

Work completed and safe working

Brief description of work carried out and any further recommendations

PPE worn and special precautions taken

Technical data and measurements

DTC code

Sensor type

Expected waveform/output

Measured output

Air gap if relevant

Wiring condition

Tone ring condition

Related systems affected

Special tools including any required calibration

Knowledge questions

1. What is the purpose of an ABS wheel speed sensor?

2. What does the term air gap mean?

3. Name causes of an illuminated ABS warning lamp.

Knowledge questions continued

4. Why can ABS faults affect other vehicle systems?

5. Why may an oscilloscope provide better information than a basic meter?

6. How would you explain an ABS fault to a customer?

7. What further tests or authorisation may be required?

Assessor observation notes

Record observations, evidence reviewed and any areas requiring further development

TASK 1: BLOCK 8 APPRENTICE ASSESSMENT

Declarations, sign-off and assessment decision

Assessment declarations

Apprentice declaration: I confirm that the work recorded in this assessment document is my own work and that I completed the task under the conditions stated.

Assessor declaration: I confirm that I have observed and/or reviewed the evidence recorded in this document and that the assessment decision is accurate.

Apprentice sign-off

Apprentice name

Date signed

Apprentice signature

Assessor sign-off

Assessor name

Assessor PIN

Date signed

Assessor signature

Sign using your PDF reader's signing tool, or sign by hand after printing.

Assessment decision and validation

Assessment decision - select Pass or Refer; tick resubmission if required

- [ ] Pass
- [ ] Refer
- [ ] Resubmission required

Assessment / evidence reference

Assessor feedback and action points